

PROFESSIONAL SUMMARY:

- Data Engineer with over 6+ years of experience in designing, building, and optimizing large-scale data pipelines and platforms across cloud (Azure, AWS) and on-premises environments. Proven expertise in modernizing and migrating data platforms to the cloud, including Spark clusters, data warehouses, and data lakes.
- Extensive experience in ETL/ELT processes, data modeling, and data integration using tools like Azure Data Factory, Databricks, Snowflake, AWS Glue, and Hadoop ecosystem (HDFS, Hive, HBase). Skilled in designing and implementing batch and real-time data processing solutions using Spark, Kafka, and Azure Stream Analytics.
- Proficient in Big Data technologies such as Hadoop, Spark, Hive, and Kafka, with hands-on experience in building scalable and efficient data pipelines for both structured and unstructured data. Strong background in data ingestion, transformation, storage, and analytics across Snowflake, Azure Synapse, and AWS Redshift.
- Expertise in data visualization and reporting using Power BI, Tableau, and SSRS, delivering actionable insights to stakeholders. Adept at data storytelling and translating complex data into clear, actionable insights for non-technical audiences.
- Strong understanding of data governance, data quality, and master data management (MDM) principles, ensuring compliance with GDPR, CCPA, and other data privacy regulations. Skilled in implementing data security measures, including encryption, access control, and data masking.
- Experienced in Agile methodologies, CI/CD pipelines, and DevOps practices for data engineering projects. Proficient in automating data pipelines using tools like Jenkins and Azure DevOps, ensuring seamless integration and deployment.
- Proven ability to optimize performance for spark applications, SQL queries, and data pipelines, ensuring scalability and cost-efficiency in cloud environments. Strong knowledge of data Lakehouse architectures and hybrid cloud solutions, enabling seamless integration of on-premises and cloud-based systems.
- Skilled in machine learning integration with data pipelines using MLib and Azure ML, enabling advanced analytics and predictive modeling. Experience in collaborating with data scientists and analysts to deliver end-to-end data solutions.
- Demonstrated success in data migration projects, including legacy system modernization and cloud adoption. Proficient in designing and implementing data lakes and data warehouses, ensuring scalability, reliability, and performance.
- A collaborative team player with excellent communication skills, capable of working with cross-functional teams to deliver data-driven solutions. Proven ability to mentor junior team members and lead data engineering initiatives, fostering a culture of innovation and continuous improvement.
- Strong troubleshooting skills, adept at resolving data pipeline issues with minimal downtime. Expertise in cost optimization for cloud data platforms, implementing efficient data storage and processing strategies to reduce operational expenses.
- Comprehensive knowledge of real-time data streaming using Kafka, Spark Streaming, and Azure Stream Analytics, enabling real-time decision-making and analytics.

TECHNICAL SKILLS:

Cloud Platforms	Azure (Data Factory, Synapse, Databricks, Data Lake, SQL DW), AWS (S3, EC2, Redshift, Glue, Lambda), Snowflake
Big Data Tools	Hadoop, Spark, Hive, Kafka, Sqoop, Flume, Pig, Oozie, Zookeeper
Databases	Azure SQL, PostgreSQL, Oracle, SQL Server, Snowflake, Cosmos DB, MongoDB
ETL/ELT Tools	Azure Data Factory, AWS Glue, SSIS, Informatica
Programming Languages	Python, Scala, SQL, Java, PySpark
Data Visualization	Power BI, Tableau, SSRS, Excel (PowerPivot, Power View)
Scripting	Shell Scripting, PowerShell, Azure CLI, AWS CLI
DevOps & CI/CD	GitHub, Azure DevOps, Jenkins, Airflow
Methodologies	Agile, Scrum, Waterfall

PROFESSIONAL EXPERIENCE:

Sr Azure Data Engineer
Fulton Bank-Lancaster, PA

Aug 2022- Present

Responsibilities:

- Designed and implemented end-to-end data pipelines using Azure Data Factory, Databricks, and Synapse Analytics to ingest, transform, and load data from multiple sources into Azure Data Lake and Azure SQL Data Warehouse, ensuring scalability and reliability.
- Spearheaded the migration of on-premises data warehouses to Azure Synapse and Snowflake, optimizing query performance and reducing operational costs by 30% through efficient resource utilization and architecture redesign.
- Developed PySpark and Spark SQL scripts for complex data transformation and aggregation tasks, enabling real-time analytics and reporting capabilities for business stakeholders.
- Implemented a Data Mesh architecture to decentralize data ownership, empowering business units to manage their data domains while improving data quality, accountability, and accessibility across the organization.
- Built robust CI/CD pipelines in Azure DevOps to automate the deployment of data pipelines, ETL workflows, and infrastructure-as-code (IaC) templates, reducing deployment errors and improving release cycles.
- Integrated Snowflake with Azure Data Factory and AWS Glue to enable seamless data ingestion and processing across multi-cloud environments, ensuring data consistency and interoperability.
- Designed and developed interactive Power BI dashboards to visualize key business metrics, enabling stakeholders to make data-driven decisions and track performance in real-time.
- Optimized Apache Spark applications by tuning batch intervals, parallelism, and memory configurations, resulting in a 40% reduction in data pipeline processing time.
- Collaborated with cross-functional teams to define and enforce data governance policies, ensuring compliance with enterprise data standards and regulatory requirements.
- Enhanced data security by implementing data encryption, role-based access controls (RBAC), and auditing mechanisms in Azure SQL Database and Azure Data Lake.
- Developed comprehensive data lineage documentation to track data flow, transformations, and dependencies across the organization, improving transparency and troubleshooting efficiency.
- Designed and implemented real-time data streaming solutions using Kafka and Azure Stream Analytics, enabling low-latency data processing for critical business applications.
- Mentored and trained junior data engineers on best practices for data engineering, including tool usage, coding standards, and performance optimization techniques.
- Conducted performance benchmarking and cost analysis for cloud data platforms, providing recommendations to optimize resource allocation and reduce cloud spend.
- Implemented data quality frameworks using Great Expectations and Azure Data Factory to validate and monitor data accuracy, completeness, and consistency across pipelines.
- Designed and deployed serverless architectures using Azure Functions and Logic Apps to automate data workflows and integrate with third-party APIs.
- Partnered with data scientists to operationalize machine learning models by integrating them into data pipelines, enabling predictive analytics and advanced insights.

Environment: Azure Data Factory, Databricks, Azure Synapse Analytics, Snowflake, Azure Data Lake, Azure SQL Data Warehouse, PySpark, Spark SQL, Azure DevOps, AWS Glue, Power BI, Apache Spark, Kafka, Azure Stream Analytics, Great Expectations, Azure Functions, Logic Apps, Python, SQL, CI/CD, Data Mesh, Data Governance, RBAC, Data Encryption, Data Lineage, Machine Learning Integration.

Azure Data Engineer
OSF Health Care- Chicago, IL

Sep 2020 – Jul 2022

Responsibilities:

- Designed and optimized scalable ETL pipelines using Azure Data Factory and Apache Airflow to ingest, process, and store structured and semi-structured data from diverse sources, ensuring efficient data flow and integration.
- Architected end-to-end data pipelines, extracting data from APIs, databases, and flat files, transforming it using PySpark and Spark SQL in Azure Databricks, and loading it into Azure Data Lake and Snowflake for analytics, reducing data processing time by 30%.
- Led the migration of on-premises data solutions to Azure Cloud (Data Lake, Synapse, and Snowflake), ensuring minimal downtime and data integrity during the transition.
- Collaborated with data scientists to integrate machine learning models into data pipelines, enabling predictive analytics and enhancing decision-making across the organization.
- Optimized data storage in Azure Data Lake using partitioning, indexing, and compression techniques, improving storage efficiency and query performance in Azure Synapse.
- Leveraged Azure Stream Analytics and Azure Event Hub for real-time streaming data processing, providing hybrid data solutions for operational and analytical use cases.
- Designed and implemented a scalable data architecture in Azure to handle high-throughput data streams, enabling efficient management of large and rapidly growing datasets.

- Enhanced data security and privacy by implementing role-based access control (RBAC) and encryption mechanisms for data at rest and in transit, ensuring compliance with HIPAA, GDPR, and CCPA regulations.
- Integrated Azure Purview to automate data cataloging and lineage tracking, improving data governance and traceability for compliance and auditing purposes.
- Built automated data pipelines feeding data into Power BI, enabling real-time analytics and self-service reporting capabilities for business stakeholders.
- Set up end-to-end monitoring for Azure Data Factory pipelines using Azure Monitor and Log Analytics, ensuring pipeline health and proactively resolving anomalies.
- Created and maintained automated CI/CD pipelines using Azure DevOps, enabling efficient deployment of code and data infrastructure changes with minimal downtime.
- Implemented cost-optimization strategies by dynamically scaling Azure resources and optimizing query execution plans in Azure Synapse, reducing monthly cloud costs by 20%.
- Developed a disaster recovery strategy using Azure Backup and Azure Site Recovery, ensuring business continuity and data resilience during system failures.
- Tuned data models and optimized complex SQL queries in Snowflake and Azure Synapse, improving query response time by 40% for large datasets.
- Collaborated with cross-functional teams, including data scientists, business analysts, and IT operations, to deliver data solutions that met business requirements and enhanced analytical capabilities.
- Integrated external data sources through REST APIs into Azure Data Factory pipelines, enabling data ingestion from third-party services for richer analytics.
- Implemented data anonymization techniques and dynamic data masking in sensitive datasets, ensuring privacy while enabling analytical processing.
- Utilized Delta Lake within Azure Databricks for enhanced data consistency and ACID transactions, ensuring high data integrity in batch and streaming workloads.
- Configured and optimized scheduled data loads within Azure Data Factory, ensuring timely and reliable updates to the data warehouse while minimizing resource consumption.
- Created dashboards and real-time visualizations in Power BI, drawing from real-time data streams and batch data, enabling executive teams to make informed decisions quickly.
- Developed robust ETL processes for data integration across multiple cloud platforms (Azure, AWS), ensuring efficient data loading and transformation into cloud-based data warehouses like Snowflake and Azure Synapse.
- Integrated real-time data ingestion with Apache Kafka and Azure Stream Analytics, enabling seamless streaming data processing for event-driven analytics.
- Implemented Azure Data Lake and Databricks for large-scale data storage and processing, optimizing both performance and cost-efficiency.

Environment: Azure Data Factory, Apache Airflow, Azure Data Lake, Snowflake, Azure Synapse, Azure Databricks, PySpark, Spark SQL, Azure Stream Analytics, Azure Event Hub, Power BI, Azure Monitor, Log Analytics, Azure DevOps, Azure Purview, Apache Kafka, Delta Lake, REST APIs, Azure Backup, Azure Site Recovery.

Hadoop Developer
Aptiv, Troy, MI

Oct 2018 – Aug 2020

Responsibilities:

- Designed, built, and optimized Hadoop-based data pipelines leveraging HDFS, MapReduce, Hive, and HBase for large-scale data processing and analytics.
- Developed and deployed Spark Streaming pipelines integrated with Kafka to enable real-time data ingestion and analytics, significantly improving the speed and accuracy of business decision-making.
- Implemented ETL processes using Hadoop Sqoop to seamlessly migrate data from relational databases (e.g., MySQL, Oracle) to HDFS, ensuring efficient data transformation for downstream analytics.
- Designed and optimized complex Hive queries to process, transform, and analyze structured data, enabling actionable insights for business reporting and analytics.
- Utilized efficient file formats such as Avro, Parquet, and ORC for optimized data storage, retrieval, and processing within Hadoop environments.
- Ensured data integrity and quality by implementing robust data validation, deduplication, and cleansing processes within the Hadoop ecosystem.
- Developed end-to-end batch and stream processing solutions using Hadoop MapReduce, HBase, and Spark to create comprehensive data processing pipelines.
- Integrated cloud storage and services (AWS S3, Azure Blob Storage) into Hadoop workflows to enable scalable, cost-efficient data processing and storage solutions.
- Containerized Hadoop-based applications using Docker, ensuring portability, scalability, and ease of deployment across diverse environments.

- Automated and orchestrated complex data workflows using Apache Oozie for scheduling, dependency management, and efficient pipeline execution.
- Implemented data governance and security policies to ensure compliance with regulatory frameworks such as GDPR and CCPA, safeguarding sensitive data.
- Developed interactive and dynamic reporting solutions using Power BI and Tableau, enabling business users to gain real-time insights into operational and financial data.
- Optimized the performance of data queries and pipelines by fine-tuning Hadoop jobs and Spark transformations, resulting in faster execution and reduced processing times.
- Integrated Git for version control and implemented automated CI/CD pipelines using Jenkins to streamline deployments and ensure consistent updates to data pipelines.
- Collaborated with cross-functional teams to gather requirements, design solutions, and deliver data-driven insights to stakeholders.
- Conducted performance benchmarking and capacity planning to ensure the Hadoop ecosystem could handle growing data volumes and processing demands.

Environment: Hadoop (HDFS, MapReduce, Hive, HBase), Spark, Kafka, Sqoop, AWS, Azure, Docker, Oozie, Power BI, Tableau, Git, Jenkins, Avro, Parquet, ORC, GDPR, CCPA.

Education: Masters in computer and Information Science from University of Texas.